

SAFETY QUIZ:CHE382-Chemical Engineering Laboratory II



Please answer the following questions using the information from the 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases given in Safety Assignment I. Note: Answers from another source may not be considered correct. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

Adil Feratovic

1. What is the definition of a toxic chemical?

A toxic chemical is a substance that causes

Any chemical that, when applied to, injected into, or created within the body at comparatively tiny levels, can cause structural damage or disruption of function.

2. Name the 4 types of toxicity damage from a toxic chemical:

-
1. Damage to structure
 2. Disturbance of function
 3. Acute toxicity
 4. Chronic toxicity
-

3. Name two parameters that can affect toxicity:

-
1. Dose
 2. Duration of exposure
-

4. What is the PEL (Permissible Exposure Limit)?

The maximum exposure that NIOSH recommends is known as the Permissible Exposure Limit, or PEL. These are eventually regulated by OSHA and are often taken from the TWA values provided by ACGIH.

5. What is the DOT (Department of Transportation) definition of a compressed gas?

University of Illinois at Chicago

Department of Chemical Engineering

According to DOT, compressed gases are classified as combustible materials with a pisa more than 40 at 100 F or nonflammable materials with a psia greater than 40 at 70 F or 104 at 130 F.

6. a) How does one determine how much material remains in a non-liquified gas cylinder?

By reading the cylinder pressure and relating it to the known cylinder volume

- b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

The high-pressure gauge

7. How does one determine how much material remains in a liquified gas cylinder?

By weighing the cylinder

8. Why do insidious hazards represent a problem to workers?

Because they are typically not seen, smelled, tasted, or felt, insidious hazards, which can potentially pose health risks, are missed and disregarded. Depending on the substance, they can have chronic effects.

9. What are the four general categories of insidious hazards?



-
1. Carcinogen
 2. Mutagen
 3. Teratogen
 4. Chronic toxicant
-

10. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

-
1. Ventilation
 2. PPE
-

University of Illinois at Chicago

Department of Chemical Engineering

11. What is the definition of a reactive chemical?

Chemicals that undergo violent reactions in which the spontaneous release of heat or gases happen too quickly for the environment to properly disperse are known as reactive chemicals.

12. Name the two groups and their sub-headings used to categorize reactive chemicals:



-
1. Water reactive chemicals
 2. Unstable chemicals
-

13. List two precautionary measures for handling and/or storing reactive chemicals.

-
1. Keep away from incompatible materials
 2. Protect from heat, moisture, and shock
-

14. How are corrosive chemicals generally defined?

These are chemicals that cause visible destruction or irreversible damage by chemical action



15. How do corrosive chemicals act on body tissues?



They destroy tissue by chemical reaction

University of Illinois at Chicago
Department of Chemical Engineering

16. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

True

17. Name the six categories of common corrosive liquids.



-
1. Inorganic acids
 2. Organic acids
 3. Alkalis
 4. Halogens
 5. Acid halides
 6. Oxidizing agents
-

18. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

Alkaline corrosive solids

19. Name the three components that make up the FIRE TRIANGLE:

-
1. Fuel
 2. Oxidizer
 3. Ignition Source
-

20. What category of flammables is the most common source of fires in industry?

Flammable solvents